

TABLE I
Instructions for Image Acquisition

Instruction	Descriptions
preview(obj)	Immediately activates a live image preview window for the video input object 'obj'.
closepreview(obj)	Closes the image preview window associated with image acquisition object 'obj'.
cellisp(c) or cellisp(dev_info, SupportedFormats)	Recursively displays the contents of a cell array
start(obj)	The control starts capturing the frames. The frames captured are stored in the memory. The control also obtains exclusive use of the image acquisition device associated with the video input object 'obj' and locks the device configuration. Starting an object is a necessary first step to acquire image data.
triggerconfig(obj, 'manual')	Sets the trigger configuration. Here, 'manual' gets the image only when the video is triggered.
set(obj, 'FramesPerTrigger', 1)	Sets the number of frames captured per trigger. Here only one frame is captured per trigger.
set(obj, 'TriggerRepeat', Inf)	Sets the number of times for which the trigger can be repeated.
flushdata(obj)	Removes all data from the data acquisition engine and resets the SamplesAvailable property to zero.
stop(obj)	Stops video capturing.
delete(obj)	Deletes the object 'obj'
clear obj	Clears the memory buffer

Note: Inf- indicates that there can be any number of triggers we want. If we specify any number then we will have to start the video capture again after we have finished triggering the specified number of times.

ter's desktop. Run the software and the screen shows the main Matlab application window containing the following sub-windows:

1. Command Window. In this window, the user types Matlab commands and expressions at '>>' prompt and the outputs of these commands are displayed.

2. Workspace browser. This window shows variables defined in the current session and also some information about them.

3. Current directory. This window displays contents of the current directory and its path.

4. Command history window. This window contains a record of the commands (including both current and previous sessions) that the user has entered in 'Command' window.

5. Others. There are one or more windows which are shown only when the user activates graphics.

Matlab has built-in adaptors to access the camera. At the Matlab command prompt, write the 'imaqhwinfo'

instruction and press 'Enter' key. This instruction returns information about all the adaptors available in the system:

```
ans =
InstalledAdaptors: {'winvideo'}
MATLABVersion: '7.0.1 (R14SP1)'
ToolboxName: 'Image Acquisition
Toolbox'
ToolboxVersion: '1.7 (R14SP1)'
```

The 'info=imaqhwinfo' instruction returns information about a specific device accessible through a particular adaptor.

The output of the 'info = imaqhwinfo('winvideo')' instruction is:

```
info =
AdaptorDllName: G:\MATLAB701\toolbox\
imaq\imaqadaptors\win32\
mwinvideo\imaq.dll'
AdaptorDllVersion: '1.7 (R14SP1)'
AdaptorName: 'winvideo'
DeviceIDs: {1x0 cell}
DeviceInfo: {1x0 struct}
```

The output of the 'dev_info = imaqhwinfo('winvideo', 1)' instruction is:

```
dev_info =
DefaultFormat: 'RGB24_640x480'
DeviceFileSupported: 0
DeviceName: 'WebCam Vista #2'
DeviceID: 1
ObjectConstructor:
'videoinput('winvideo', 1)'
SupportedFormats: {1x11 cell}
```

Again, the 'obj = videoinput('adaptorname', deviceID, 'format')' instruction constructs a video input object where the 'adaptorname' string specifies the name of the device adaptor that the object 'obj' is associated with. 'deviceID' is the identifier of the device in numerical number. If 'deviceID' is not specified, the first available device ID is used. The 'format' string specifies the video format for the object. If the format is not specified, the device default format is used. For example, the output of the 'obj=videoinput('winvideo', 1, 'RGB24_640x480')' instruction will be:

```
Summary of Video Input Object Using
'WebCam Vista'.
Acquisition Source(s): input1 is
available.
Acquisition Parameters: 'input1' is
the current selected source.
10 frames per trigger using the
selected source.
'RGB24_640x480' video data to be
logged upon START.
Grabbing first of every 1 frame(s).
Log data to 'memory' on trigger.
Trigger Parameters: 1 'immediate'
trigger(s) on START.
Status: Waiting for START.
0 frames acquired since starting.
0 frames available for GETDATA.
```

Binary images (B&W). A binary image is a logical array of 0's and 1's. Numeric-array images consisting of 0's and 1's are converted into binary, with '1' indicating white colour (maximum intensity) and '0' indicating black colour (minimum intensity):

```
>>imBw=[1 0 1; 0 1; 1 1 0]
```

RGB images. A red, green and blue (RGB) image is an MxNx3 array of colour pixels, where each colour